AMERICAN TERAWATT

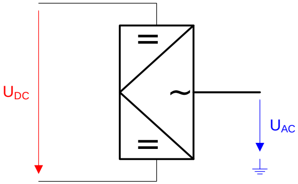
HVDC Design

American Terawatt | July 23, 2026

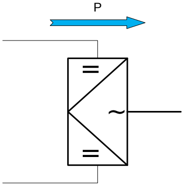
DC Power Transmission

Converters, topologies and load flow

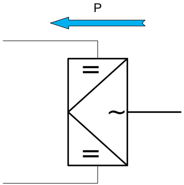
Converter



Inverter

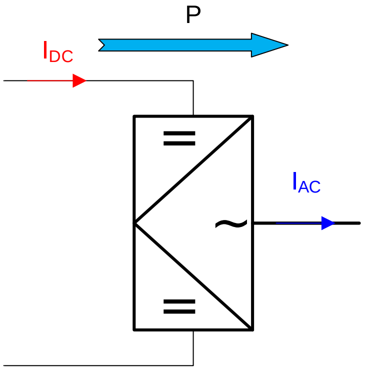


Rectifier

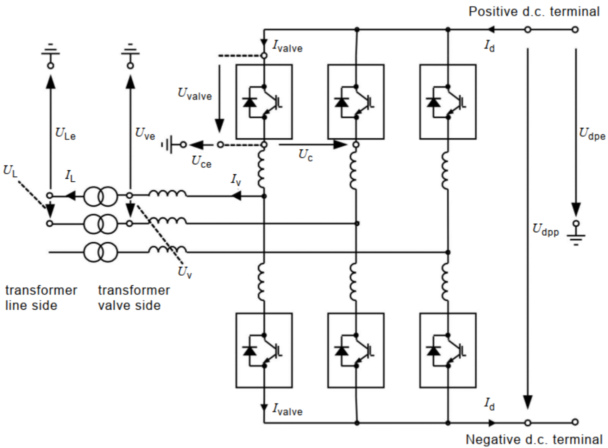


Converter - Load flow definition

- Positive load is defined for the inverter case (generator convention).
- In the USA a common term is "Inverter-Based Resources" (IBR).
- IBR are not only HVDC systems but all kinds of generators:
  - Wind Turbine Generators (WTG)
  - PV, . . .
- A VSC-HVDC system can be an inverter or a rectifier depending on the power-flow direction; some applications require changing direction.



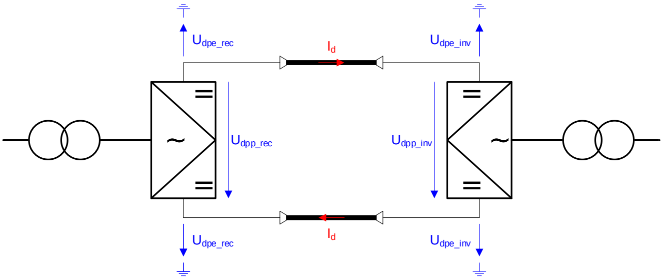
Converter terminology



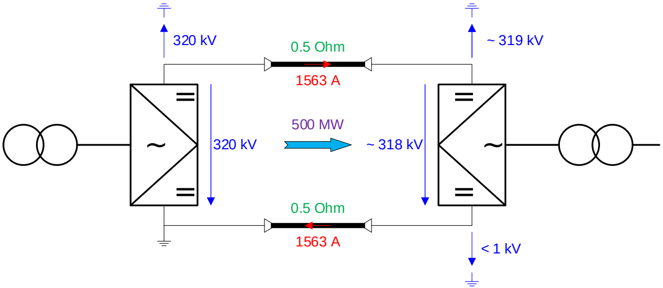
IEC 62747:2014

DC load flow definitions

- DC equipment (incl. DC line) voltage is defined by the voltages U_{dpe_rec} and U_{dpe_inv}
- Converter size is defined by U_{dpp_rec}
- HVDC systems have bidirectional transmission capability, which means that every individual converter can be either rectifier or inverter



Monopole



DC load flow

$$U_{dpp\_rec} = U_{dpe\_rec}$$

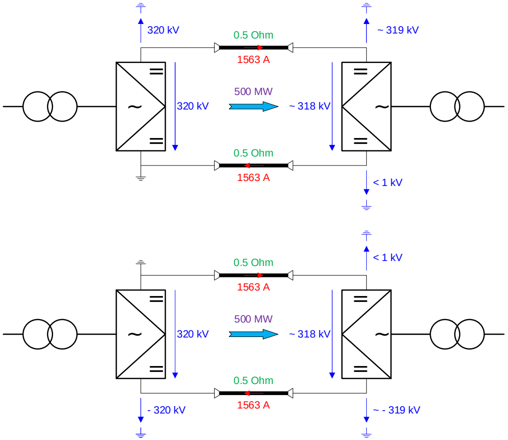
$$I_{DC} = \frac{U_{dpp\_rec} - U_{dpp\_inv}}{2 \cdot R_{DC}}$$

$$U_{dne\_rec} = I_d \cdot R_{DC}$$

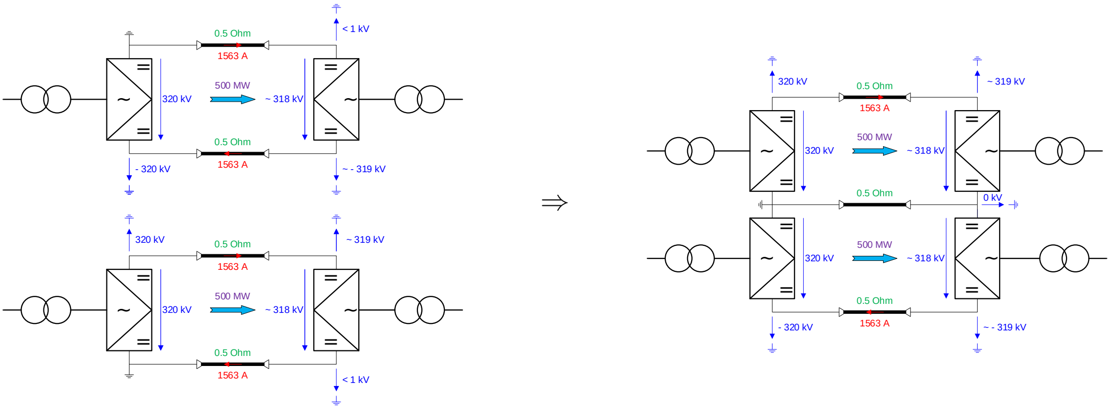
- Max. voltage created by converter: U_{dpp_rec}
- Max. equipment voltage:
- DC cables have different voltage ratings

Monopole - extending power transmission

- **Higher DC voltage**
  - Limited equipment availability
  - No redundancy
- **Higher DC current**
  - Higher DC line costs
  - No redundancy
- **Additional monopole**
  - Additional converter equipment
  - 4 DC conductors
  - Increased reliability

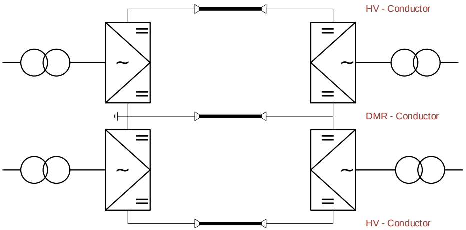


Monopole → Bipole (BIP)

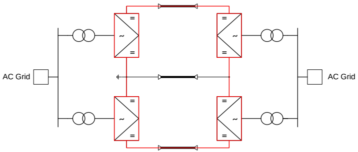


Bipole (BIP)

- 3 conductors of 2 different types.
- Two poles that can transmit power independently.
- On a fault in one pole, the other pole still transmits power — redundancy.
- In symmetrical operation (both poles equal), no current flows through the DMR.

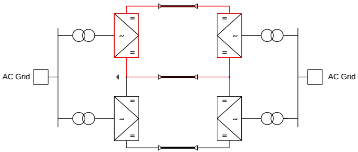


BIP operation modes



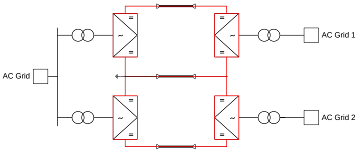
Bipolar — balanced

- Equal power for poles
- No DMR current
- Loss-optimized



Monopolar

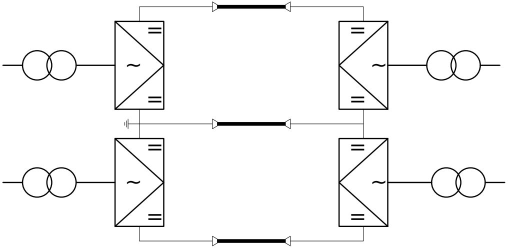
- Emergency mode
- System-design definition

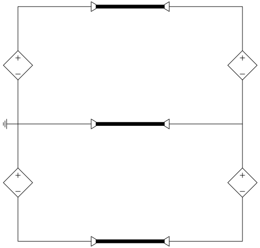


Bipolar — unbalanced

- Split-busbar operation
- DMR current flow

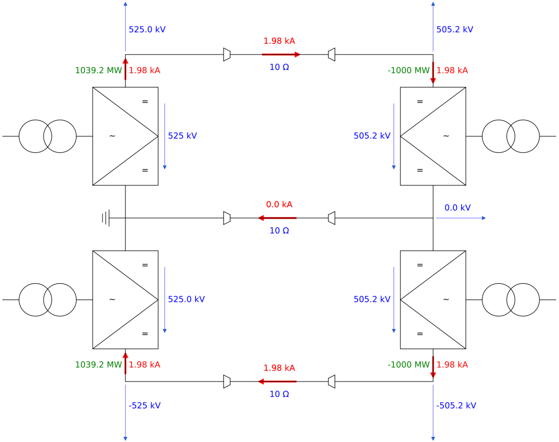
DC load flow



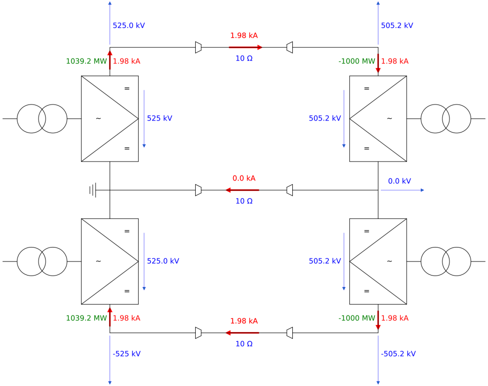


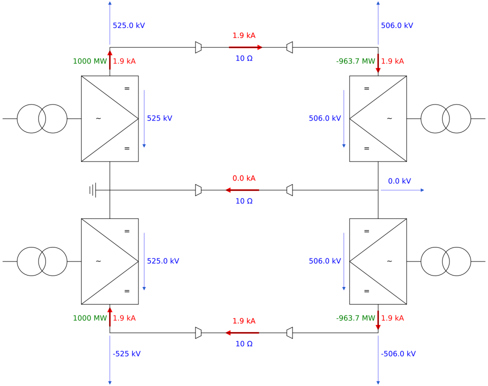
For DC load-flow calculation the converter can be represented as an ideal voltage source.

Balanced operation



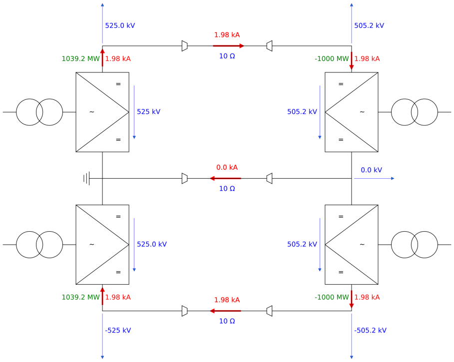
Balanced operation - power definition



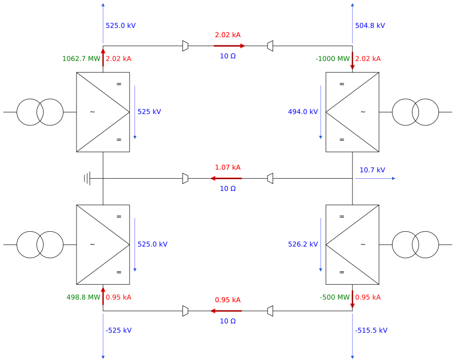


Balanced vs. unbalanced operation

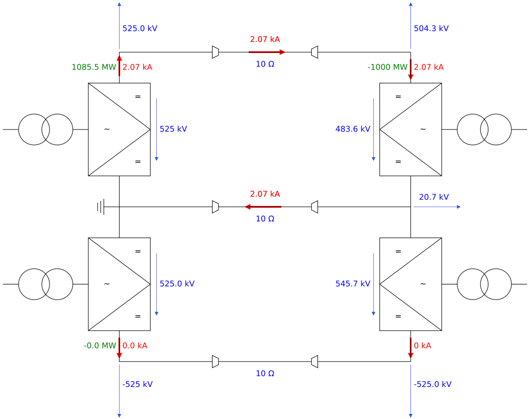
Balanced mode



Unbalanced mode

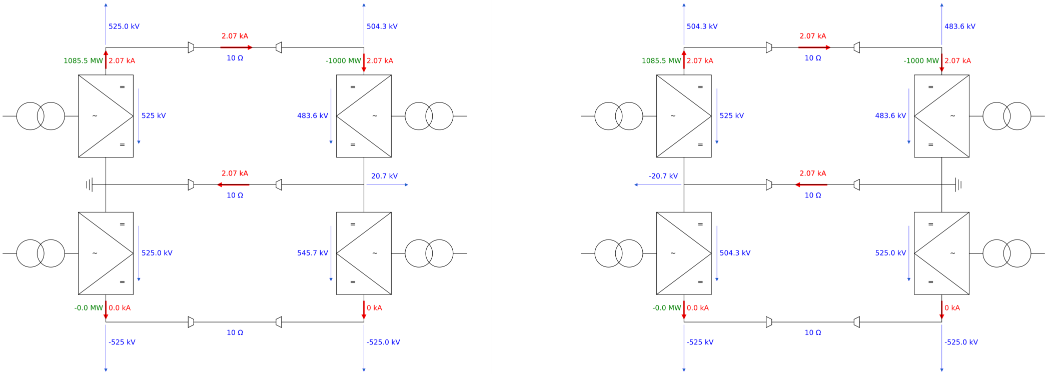


Monopolar operation



- Usually for a short period, until system reconfiguration is finished.
- For longer operation, the DMR must be designed accordingly.

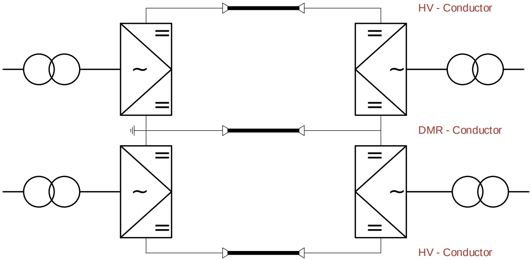
System earthing



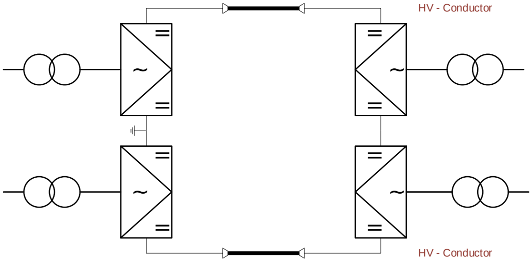
Earthing of the BIP system has no impact on the resulting currents and losses.

BIP → Rigid BIP

Bipole

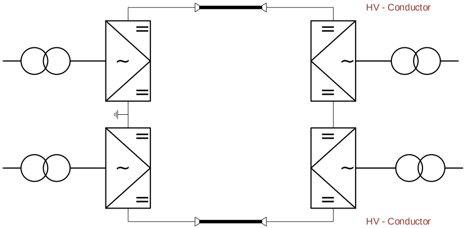


Rigid Bipole

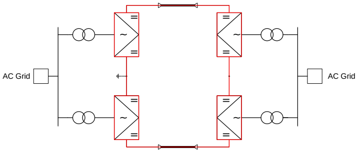


Rigid BIP

- 2 conductors of 1 type.
- Both poles must transmit the same power (same current flows through both poles).
- On a fault in one pole, the other pole might not be able to transmit the power.

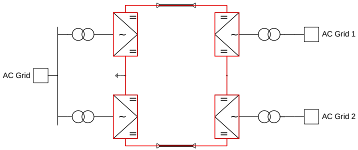


Rigid BIP operation modes



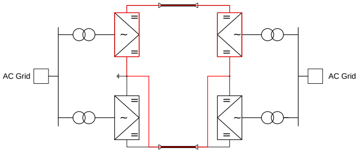
Common-busbar

- Standard operation mode.



Split-busbar

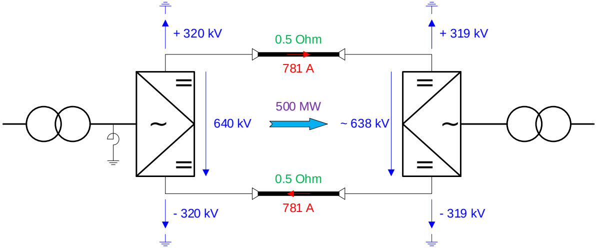
- Poles must keep equal power in all modes (incl. dynamic events)



Monopolar

- Reconfiguration takes > 1 s.

Symmetrical Monopole (SMP)



- High-impedance earth reference on the AC side
- $U_{dpp\_Rec} = 2 \cdot |U_{dpe\_Rec}|$
- $I_d = \frac{U_{dpp\_rec} - U_{dpp\_inv}}{2 \cdot R_d}$
- Only for VSC-HVDC systems
- DC cables have the same voltage rating

Modular Multilevel Converter (MMC)

Power transmission parameters:

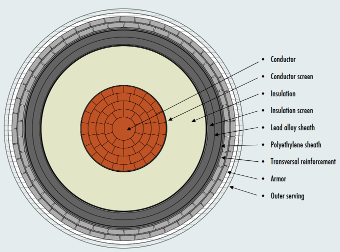
- Power: 1000 MW
- DC voltage: $\pm$ 320 kV
- Max. DC current: 1563 A

	2 $\times$ Monopole	**1 $\times$ BIP**	**1 $\times$ SMP**
Transformers	$--$	$--$	$++$
DC lines	4 lines / 2 types	3 lines / 2 types	2 lines / 1 type
Converters	4 converters	4 converters	2 converters
Converter rating (U_{dpp})	320 kV	320 kV	640 kV
Losses	$--$	$-$	$+$
Reliability	$++$	$+$	$--$

DC cable

- **Insulator** — XLPE (cross-linked polyethylene) or MI (mass-impregnated).
- **Conductor** — copper or aluminium.
- Balance between DC voltage and DC current for the most economical design.
- Insulator is generally cheaper than conductor $\rightarrow$ higher voltage rating is preferred

Project voltages: 80/200/250/**320**/400/**525** kV (525 kV is new standard for ≥ 2000 MW).



www.borealisgroup.com

Topology comparison

	Sym. Monopole	Rigid Bipole	Bipole w/ MR
DC conductors	2	2	3
Power on converter fault	0%	50%	50%
Power on DC conductor fault	0%	0%	50%
Economical DC voltage	≤ 400 kV	> 500 kV	> 500 kV
Costs	$	$$	$$$
Max. power	1400 MW (320 kV) 1500 MW (400 kV)	2000 MW	3000 MW